

JOHNSON MUGARRA

Data Scientist, Research and Data Consultant, Technical Lead Consultant

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PROFESSIONAL SUMMARY

Data scientist and research consultant with an M.Sc. in Data Science and five years turning field and operational data into decisions across East Africa. At Enimiro I design studies - child labour, living income, pollination - and build the pipelines, dashboards, and yield models behind them. At the Center for Research and Communication I coordinate proposals and provide technical leadership for evaluations covering design, sampling, tools, data quality, analysis, and reporting. Earlier roles cover vanilla value-chain forecasting with Makerere University and MAAIF, process evaluation at IPA, and field research for USAID SITES.

CORE SKILLS

- Statistical software: Stata, R, Python, SQL, SAS, SPSS, Excel
- Research and evaluation: survey design, piloting and translation, computer-assisted interviewing, key informant interviews, focus group discussions, transcription
- Data collection and quality: questionnaires, tablet-based collection, enumerator training and supervision, validation, back-up and documentation
- Visualization and reporting: Power BI, Tableau, dashboards, donor-ready reports
- Data systems: SurveyCTO API, PostgreSQL, Google Sheets, ETL pipelines, GeoPandas
- Machine learning and AI: scikit-learn, time-series forecasting, neural networks and CNNs in TensorFlow and PyTorch, and daily AI tooling for cleaning, code, and drafting with outputs verified by hand
- Proposal development: concept note, strategy, design, methodologies and work plans, budgeting, quality assurance

WORK EXPERIENCE

Research and Data Consultant, November 2025 - September 2026

Enimiro Products Uganda Ltd, Kayunga, Uganda

- Coordinated with the data team to collect, clean, and organize operational data on farmer activities, crop production, sales, and sustainability programs.
- Maintained a Python pipeline moving SurveyCTO submissions into PostgreSQL raw, clean, and analysis tables, with scheduled exports to Google Sheets for dashboards.
- Designed studies end to end - proposal, methodology, survey instruments, and analysis - including the 2026 child labour study, the vanilla living income study against Anker benchmarks, pollination and maturity data modules, and the internal M&E measurement plan.
- Matched GPS records to district boundaries and maintained the farmer master with merge and change logs for audits.
- Built and maintained dashboards and reports tracking performance indicators for management and the board.
- Ran statistical and trend checks to flag risks and opportunities and translated findings into training manuals and field guidance.
- Supported crop and supply-chain research, updated survey tools, trained staff on data use, and documented methods for donor and certification reporting.

Technical Lead Consultant, January 2025 - June 2026

Center for Research and Communication (CRC), Remote

- Reviewed client requirements and evaluation criteria to shape proposal strategy and coordinated specialist inputs against deadlines.
- Wrote and edited technical sections, including methodologies and work plans, and worked with finance to align technical approaches with compliant budgets.
- Provided technical leadership for the IRC MERP endline evaluation, covering design and tools, analysis and synthesis, and integration of qualitative and quantitative findings against the results framework.
- Led quantitative inputs for IRC baseline and perception studies: sampling, instrument testing, data-quality checks, and statistical analysis.
- Served as data and M&E lead for UNICEF behavior-change and awareness studies: tools, trackers, datasets, reach analysis, and report drafting.

Researcher / Data Scientist, Vanilla Value-Chain Research, January 2024 - June 2025

Makerere University with MAAIF, Catholic Relief Services, and private sector partners, Uganda

- Researched vanilla maturity and harvest timing for national harvest-declaration considerations, combining field and crop work with statistical modelling and thermal-time analysis.
- Integrated field, satellite climate, and survey data and built time-series models to compare maturity-assessment methods and their timing, quality, and price implications.

- Conducted farmer and stakeholder interviews, field measurements, and seasonal maturity assessments across vanilla-growing districts.
- Helped establish the Kabanyolo demonstration plot and prepared models, scenario notes, and technical inputs for supervisors and stakeholders. Work concluded with program funding closure.

Research Associate, February 2023 - November 2023

Innovations for Poverty Action (IPA), Kampala, Uganda

- Supported development and implementation of process evaluation plans with program managers.
- Designed survey instruments and supported piloting, translation, and computer-assisted interviewing.
- Coordinated evaluation logistics including timelines, field staffing, and procurement, and managed data entry, cleaning, and analysis.
- Applied data-quality protocols including enumerator audits and monitoring of data collection.
- Maintained partner relationships, contributed to stakeholder meetings and reports, and supported research approvals and compliance.

Qualitative Research Associate, July 2023 - March 2024

Social and Scientific Systems for USAID Strategic Information Technical Support (SITES), Uganda

- Trained on field methods and tools and managed questionnaires and tablets for fieldwork across Eastern and Northern Uganda.
- Assessed facility readiness for Health Information Exchange and paperless viral load requests, reviewing UgandaEMR records against ART registers and viral load registers.
- Conducted key informant interviews and FGDs with clinical and data staff and checked data quality across HMIS tools and the EMR.
- Documented practical constraints affecting EMR use such as DHIS2 knowledge, internet and power stability, and computer performance, and shared facility-level feedback at each visit.
- Compiled progress and activity reports with safe back-up, storage, and transmission, and completed a 12-day assessment from 18 to 29 September 2023 within a seven-region exercise covering Kigezi, Ankole, Lango, Acholi, Busoga, Bugisu, and Karamoja.

Research Associate, September 2021 - January 2023

Bronkar (U) Limited, Kampala, Uganda

- Developed assignment workplans and tracked deliverables across research assignments.
- Designed methodologies and survey tools for qualitative and quantitative collection.
- Trained enumerators and participated in field research including interviews and structured surveys.
- Cleaned and analyzed survey data and prepared insights and conclusions for client reports and proposals.

PORTFOLIO

- GitHub and portfolio: github.com/mugarrajohnson and johnsonmugarra.com - repositories and contribution history, project write-ups, live D3.js analytics, and publications with DOIs.
- Vanilla forecasting: Prophet and ARIMA models combining NASA POWER climate data with field measurements; interactive harvest model on the website, presented nationally at Makerere in April 2025.
- Farm data platform (Enimiro): seven scheduled Python and R pipelines moving SurveyCTO submissions into PostgreSQL, feeding Power BI and Looker dashboards; private, walk-through on request.
- Automated evaluation pipeline: Python ingestion, cleaning, and report generation turning recurring evaluation cycles into one rerunnable run.

EDUCATION

M.Sc. Data Science, September 2022 - October 2024

Manipal Academy of Higher Education (MAHE), Karnataka, India

Coursework: Machine Learning, Deep Learning, Predictive Analytics, Time-Series Forecasting

B.Sc. Agricultural Science, September 2016 - May 2021

Makerere University, Kampala, Uganda

CERTIFICATIONS

SAS Certified Statistical Business Analyst
 Microsoft Certified: Power BI Data Analyst Associate
 Tableau Desktop Specialist
 Google Project Management Certificate
 Human Subjects Research (CITI)

REFERENCE

- Available on request.